

Resummation of QCD large infrared logarithms for Thrust distribution in e^+e^- annihilation

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July 22, 2024



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FACOLTÀ DI SCIENZE E TECNOLOGIE

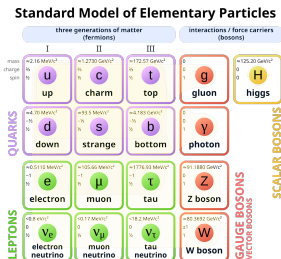
The Standard Model of Particle Physics and QCD

Collider Physics:

- Probing the fundamental structure of matter;
- Test of the Standard Model;

The Standard Model:

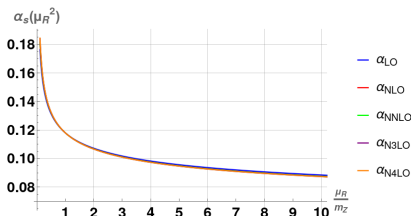
- A quantum field theory (QFT);
- Describes fundamental particles and interactions;



Quantum Chromodynamics (QCD):

- Theory of the strong interaction;
- Running coupling constant α_s ;
- $\alpha_s(m_Z^2) = 0.1180 \pm 0.0009$ (PDG 2023 average);
- Renormalization Group Equation

$$\mu_R^2 \frac{d\alpha_s}{d\mu_R^2} = -\alpha_s^2 \sum_{i=0}^{\infty} b_i \alpha_s^i;$$
- High energy $\rightarrow$ **asymptotic freedom**;
- Low energy $\rightarrow$ **confinement**;



Thrust: Definition

$$T = \max_{\hat{n}} \frac{\sum_{i=1}^N |\vec{p}_i \cdot \hat{n}|}{\sum_{i=1}^N |\vec{p}_i|} = 1 - \tau$$

- T : Thrust
- p_i : Momentum of the i -th particle
- $\hat{n}$: Thrust axis, the direction that maximizes the thrust
- Infrared-safe observable: soft emission contribution drop out and collinear splitting doesn't affect the thrust value
- Thrust value ranges from 0.5 (isotropic) to 1 (pencil like)

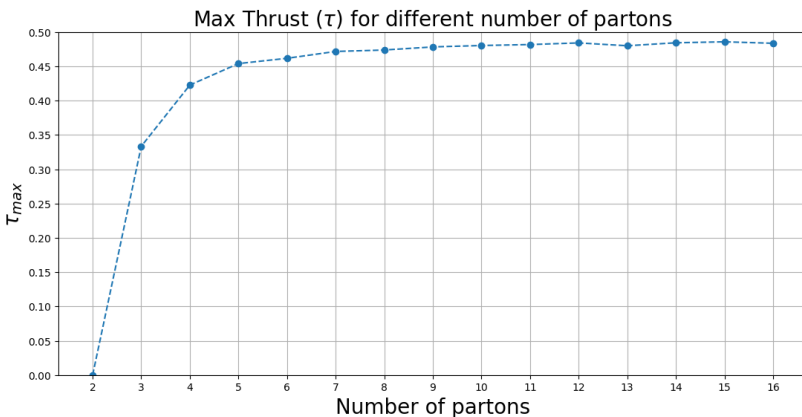


Figure 1: The maximum value of thrust τ as a function of the number of partons N in the final state. Optimization problem solved using Genetic Algorithm.

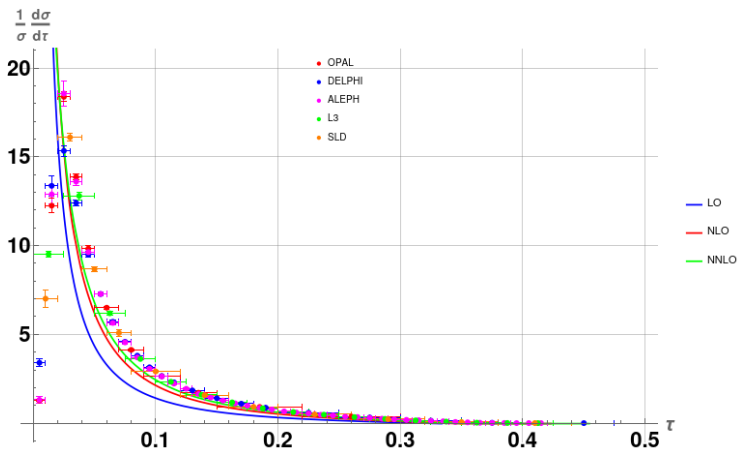


Figure 2: Fixed Order calculations compared to experimental data.

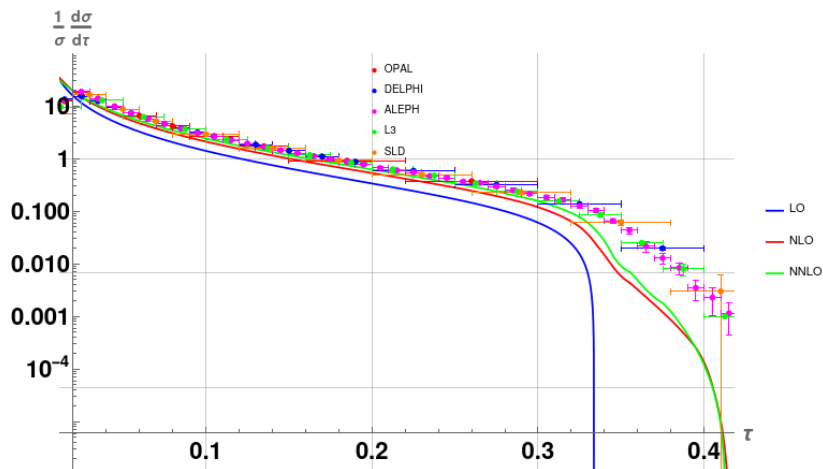


Figure 3: Log Plot of the Fixed Order calculations.

Thrust distribution

The Fixed Order expansion of the thrust distribution:

$$\begin{aligned}
 \frac{1}{\sigma_0} \frac{d\sigma}{d\tau}(\tau, \mu) &= \delta(\tau) + \overbrace{\bar{\alpha}_s(\mu) \frac{dA}{d\tau}}^{\text{LO}} + \overbrace{\bar{\alpha}_s^2(\mu) \frac{dB}{d\tau}}^{\text{NLO}} + \overbrace{\bar{\alpha}_s^3(\mu) \frac{dC}{d\tau}}^{\text{NNLO}} + \mathcal{O}(\bar{\alpha}_s^4), \\
 \frac{dA}{d\tau}(\tau) &= \frac{4}{3} \left(9\tau - \frac{3}{\tau} + \left(-6 + \frac{4}{(1-\tau)\tau} \right) \ln \left(\frac{1-2\tau}{\tau} \right) + 6 \right), \\
 \frac{1}{\sigma_0} \frac{d\sigma}{d\tau} \stackrel{\tau \rightarrow 0}{=} &\delta(\tau) + \frac{2\alpha_s}{3\pi} \left[\frac{-4 \ln \tau - 3}{\tau} + \dots \right]_+, \\
 R_T(\tau) &= \int_0^\tau \frac{1}{\sigma_0} \frac{d\sigma}{d\tau'} d\tau' = 1 + \frac{2\alpha_s}{3\pi} \left[-2 \ln^2 \frac{1}{\tau} + 3 \ln \frac{1}{\tau} + \dots \right].
 \end{aligned}$$

- $G_{nm} \bar{\alpha}_s^n \ln^m \frac{1}{\tau}$, with $m \leq 2n$ plagues the fixed order expansion;
- **Infrared Origin**: unbalanced cancellation of soft/collinear real and virtual contributions.

Resummed distribution

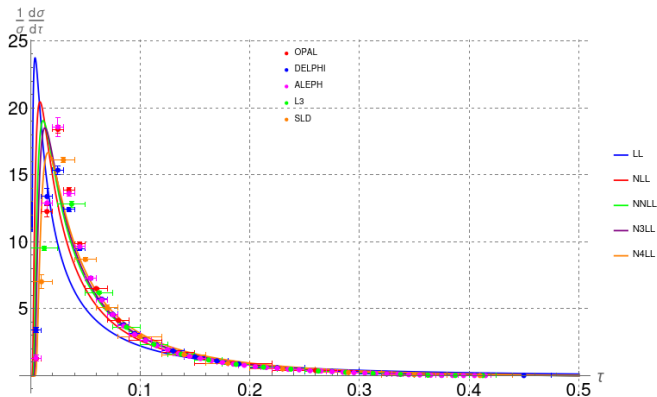


Figure 4: Resummed Distribution

Resummed distribution

$$\begin{aligned}
 R_T(\tau, \mu) &\stackrel{\tau \rightarrow 0}{\sim} \left(1 + \sum_{n=1}^{\infty} C_n \bar{\alpha}_s^n \right) \exp \left[L g_1(\lambda) + \sum_{i=2} g_i(\lambda) \alpha_s^{i-2} \right] \\
 &= \left(1 + \sum_{n=1}^{\infty} C_n \bar{\alpha}_s^n \right) \exp \left[\sum_{n=1}^{\infty} \bar{\alpha}_s^n \sum_{m=1}^{2n} G_{nm} \ln^m \frac{1}{\tau} \right], \\
 &= 1 + R^{(1)}(\tau) \bar{\alpha}_s + R^{(2)}(\tau) \bar{\alpha}_s^2 + R^{(3)}(\tau) \bar{\alpha}_s^3 + \dots, \\
 R^{(n)}(\tau) &= \sum_{m=1}^{2n} R_{nm} \ln^m \frac{1}{\tau} + C_n(\tau).
 \end{aligned}$$

		LL	NLL	NNLL	N ³ LL	N ⁴ LL
LO	$\bar{\alpha}_s^1$	G_{12}	G_{11}	C_1	-	-
NLO	$\bar{\alpha}_s^2$	G_{23}	G_{22}	G_{21}	C_2	-
NNLO	$\bar{\alpha}_s^3$	G_{34}	G_{33}	G_{32}	G_{31}	C_3
N ³ LO	$\bar{\alpha}_s^4$	G_{45}	G_{44}	G_{43}	G_{42}	G_{41}

Resummed distribution LogPlot

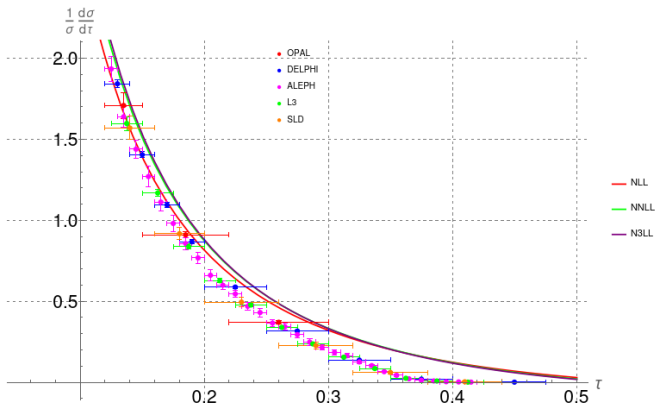


Figure 5: Resummed Distribution LogPlot

Matched distribution

$$\begin{aligned}
 R_T(\tau, \mu) &= \left(1 + \sum_{n=1}^{\infty} C_n \bar{\alpha}_s^n\right) \exp \left[L g_1(\lambda) + \sum_{i=2} g_i(\lambda) \alpha_s^{i-2} \right] + D(\tau, \alpha_s(Q^2)) \\
 &= \left(1 + \sum_{n=1}^{\infty} C_n \bar{\alpha}_s^n\right) \exp \left[\sum_{n=1}^{\infty} \bar{\alpha}_s^n \sum_{m=1}^{2n} G_{nm} \ln^m \frac{1}{\tau} \right] + D(\tau, \alpha_s(Q^2)), \\
 &= 1 + R^{(1)}(\tau) \bar{\alpha}_s + R^{(2)}(\tau) \bar{\alpha}_s^2 + R^{(3)}(\tau) \bar{\alpha}_s^3 + \dots + D(\tau, \alpha_s(Q^2)),
 \end{aligned}$$

$$R^{(n)}(\tau) = \sum_{m=1}^{2n} R_{nm} \ln^m \frac{1}{\tau} + C_n$$

$$D_1(\tau) = A(\tau) - R^{(1)}(\tau),$$

$$D_2(\tau) = B(\tau) - R^{(2)}(\tau),$$

$$D_3(\tau) = C(\tau) - R^{(3)}(\tau).$$

Matched distribution

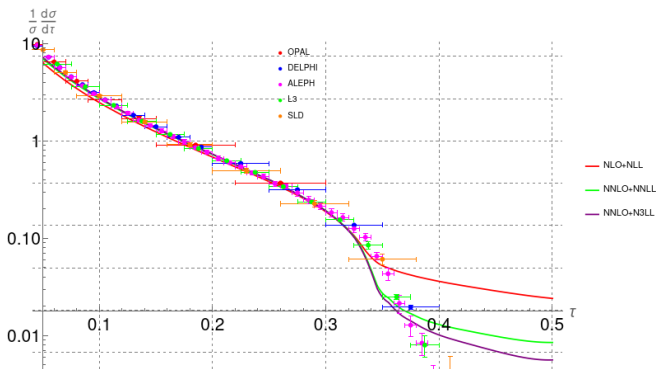


Figure 6: Matched Distribution LogPlot

Matched distribution

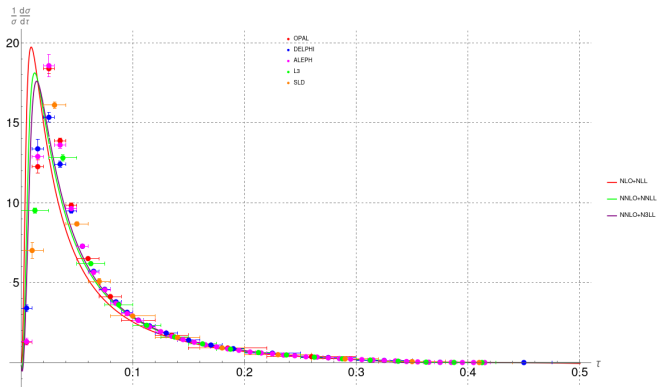
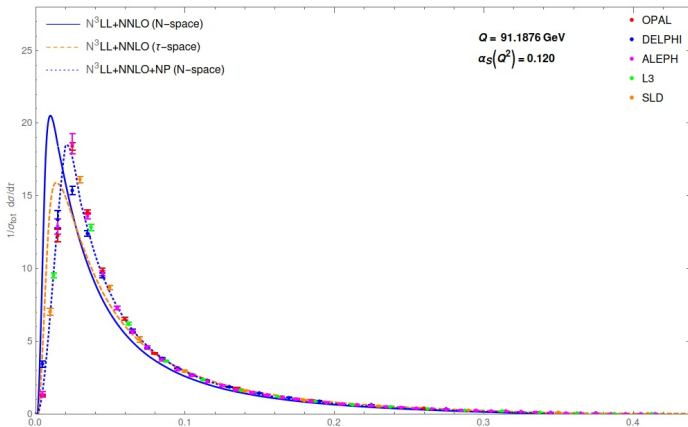


Figure 7: Matched Distribution

Preliminary Results



Summary and Future Work

In this work I have:

- Calculated resummation functions $f_i(\lambda)$ in Laplace space, ensuring consistency and extending to N⁴LL accuracy;
- Performed inverse Laplace transform for resummed distribution;
- Matched resummed expression to NNLO fixed-order calculations using the R-matching scheme;
- Achieved NNLO+N³LL accuracy for Thrust distribution predictions.

Future developments:

- Integrate phenomenological model for hadronization effects;
- Conduct fitting analysis with experimental data for accurate α_s determination;
- Investigate reliability of analytical Laplace inversion for higher orders;
- Address discrepancies in peaks of resummed Thrust distribution compared to experimental data.

Thank You!

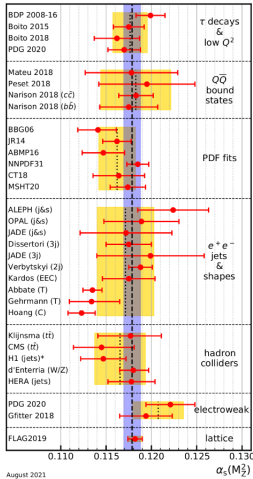


Figure 8: Particle Data Group 2021 average of α_s

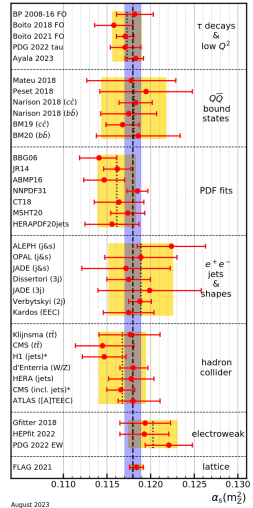


Figure 9: Particle Data Group 2023 average of α_s

How to find the Thrust axis

Given a set of N partons in the final state with momenta $\vec{p}_i$, the optimization problem for the thrust can be reduced to a combinatorial problem:

$$\vec{p}_j \cdot \vec{n} = s_j |\vec{p}_j \cdot \vec{n}| \quad \text{with } s_j = \pm 1 \quad ,$$

$$T = \max_{\vec{n}} \frac{\sum_j |\vec{p}_j \cdot \vec{n}|}{\sum_j |\vec{p}_j|} = \max_{\vec{n}} \frac{\sum_j s_j \vec{p}_j \cdot \vec{n}}{\sum_j |\vec{p}_j|} = \max_{\vec{n}} \frac{\vec{P} \cdot \vec{n}}{\sum_i |\vec{p}_i|}$$

$$\vec{P} = \sum_j s_j \vec{p}_j.$$

- 2^N possible configurations for the s_j ;
- Thrust invariant under $\vec{n} \rightarrow -\vec{n}$ transformation, $\rightarrow 2^{N-1}$ configurations;
- The configuration of all 1's and -1's doesn't respect the momentum conservation, $\rightarrow 2^{N-1} - 1$ configurations to check;
- Check the $2^{N-1} - 1$ configurations to find $\vec{P}$
- Thrust axis is $\vec{n} = \frac{\vec{P}}{|\vec{P}|}$

Minimum kinematical configuration of Thrust for 3 partons $T_{\min} = 0.6666667$ $\tau_{\max} = 0.3333333$

● Vector 1
 ● Vector 2
 ● Vector 3
 Angle 1-2: 120.00°
 Angle 1-3: 120.00°
 Angle 2-3: 120.00°

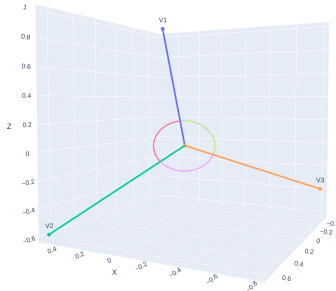


Figure 10: Kinematic configuration of the three partons in the final state that minimizes the thrust.

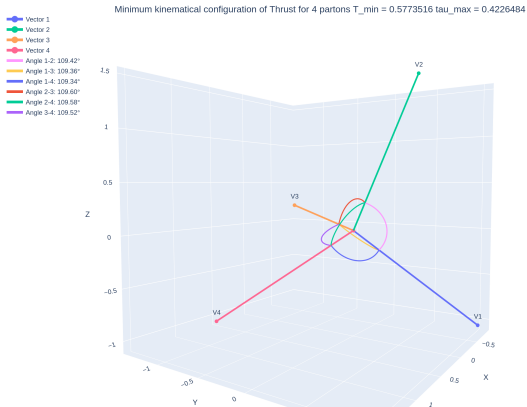


Figure 11: Kinematic configuration of the four partons in the final state that minimizes the thrust.

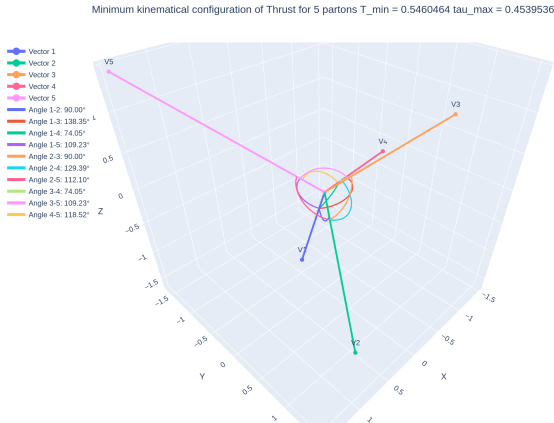
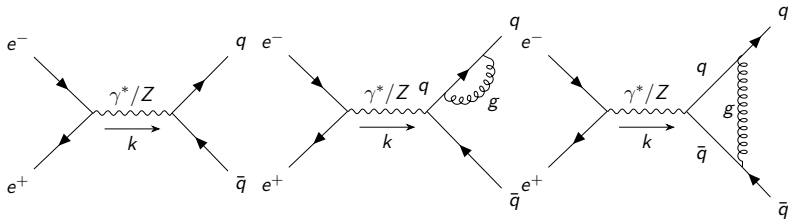


Figure 12: Approximate Kinematic configuration of the five partons in the final state that minimizes the thrust.

Real and Virtual Contributions

- Infrared Origin:** unbalanced cancellation of soft/collinear real and virtual contributions.



Infrared-safety

$$T = \max_{\hat{n}} \frac{\sum_{i=1}^N |\vec{p}_i \cdot \hat{n}|}{\sum_{i=1}^N |\vec{p}_i|} = 1 - \tau$$

- Infrared-safe observable: soft emission contribution drop out and collinear splitting doesn't affect the thrust value

$$\begin{aligned} |(1-\lambda)\vec{p}_i \cdot \vec{n}| + |\lambda\vec{p}_i \cdot \vec{n}| &= (1-\lambda)|\vec{p}_i \cdot \vec{n}| + \lambda|\vec{p}_i \cdot \vec{n}| = |\vec{p}_i \cdot \vec{n}|, \\ |(1-\lambda)\vec{p}_i| + |\lambda\vec{p}_i| &= (1-\lambda)|\vec{p}_i| + \lambda|\vec{p}_i| = |\vec{p}_i|. \end{aligned}$$

Calculations

$$\ln \tilde{J}_\nu^q(Q^2) = - \int_{N_0/N}^1 \frac{du}{u} \left[\int_{u^2 Q^2}^{u Q^2} \frac{1}{q^2} A(\alpha_s(q^2)) dq^2 + \frac{1}{2} B(\alpha_s(u Q^2)) \right] \\ + \ln \tilde{C} \left(\alpha_s(\mu^2), \frac{\mu^2}{Q^2} \right),$$

$$A(\alpha_s) = \sum_{n=1}^{\infty} A_n \left(\frac{\alpha_s}{\pi} \right)^n,$$

$$B(\alpha_s) = \sum_{n=1}^{\infty} B_n \left(\frac{\alpha_s}{\pi} \right)^n,$$

$$\ln \tilde{J}_\nu^q(Q^2) = \ln N f_1(\lambda) + f_2(\lambda) + \alpha_s f_3(\lambda) + \alpha_s^2 f_4(\lambda) + \alpha_s^3 f_5(\lambda) + \mathcal{O}(\alpha_s^n \ln^{n-4} N) \\ + \ln \tilde{C} \left(\alpha_s(\mu^2), \frac{\mu^2}{Q^2} \right).$$

Inverse Laplace Transform

$$J^q(Q^2, k^2) = \frac{1}{2\pi i} \lim_{T \rightarrow \infty} \int_{C-iT}^{C+iT} d\nu e^{\nu k^2} \tilde{J}_\nu^q(Q^2),$$

$$R^q(w) = \frac{1}{2\pi i} \int_C \frac{du}{u} e^u e^{\mathcal{F}(\alpha_s, \ln u + L)}$$

$$\stackrel{\text{Taylor}}{=} \int_C \frac{du}{2\pi i} e^{u - \ln u} e^{\mathcal{F}(\alpha_s, L) + \sum_{n=1}^{\infty} \frac{\mathcal{F}^{(n)}(\alpha_s, L)}{n!} \ln^n u}$$

$$= e^{\mathcal{F}(\alpha_s, L)} \int_C \frac{du}{2\pi i} e^{u - \ln u} e^{\sum_{n=1}^{\infty} \frac{\mathcal{F}^{(n)}(\alpha_s, L)}{n!} \ln^n u},$$

$$\begin{aligned} \ln R_T^q(\tau) &= \ln \frac{1}{\tau} g_1(\lambda) + g_2(\lambda) + \alpha_s g_3(\lambda) + \alpha_s^2 g_4(\lambda) + \alpha_s^3 g_5(\lambda) + \mathcal{O}\left(\alpha_s^n \ln^{n-4} \frac{1}{\tau}\right) \\ &+ \ln C\left(\alpha_s(\mu^2), \frac{\mu^2}{Q^2}\right). \end{aligned}$$